

Predicting Student Dropouts

Using academic, demographic, and socioeconomic data to flag at-risk students early enough to help them.

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Student Dropout Statistics



Alabama



Oregon



California

Most dropouts were predictable and preventable.

By the time a student formally drops out, the warning signs have been visible for months.

1 in 3

students in our dataset did not graduate (32% dropout).

The cost

- Lost tuition
- wasted financial aid

Must focus on early support, which is much cheaper than a lost student

The Question We Set out to Answer

"How can an institution identify students who may need support early enough to improve student outcomes?"

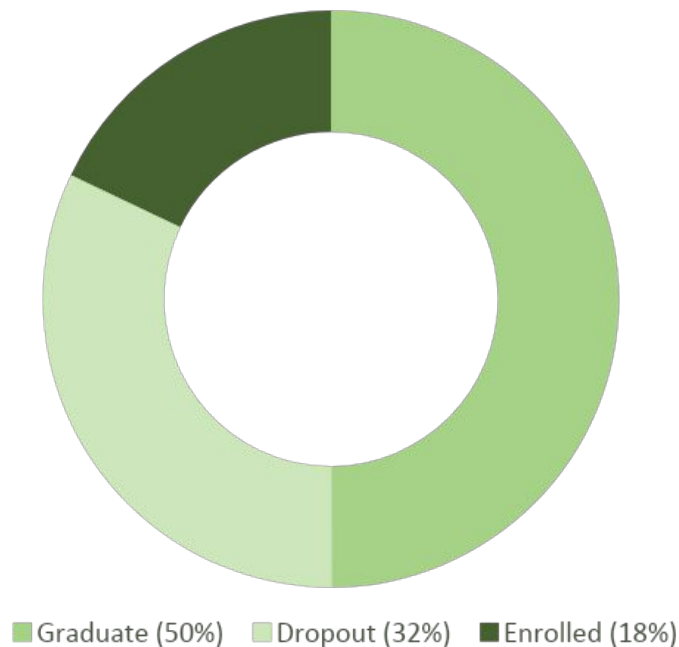
WHAT WE BUILT

A model that sorts each student into Dropout, Enrolled, or Graduate from their first-year data and flags who needs help.

WHO IT'S FOR

A university's student-success / advising team, including the people who run tutoring, check-ins, and interventions each term.

Exploratory Data Analysis



What's in each student:

Academic: grades, classes passed/failed each semester

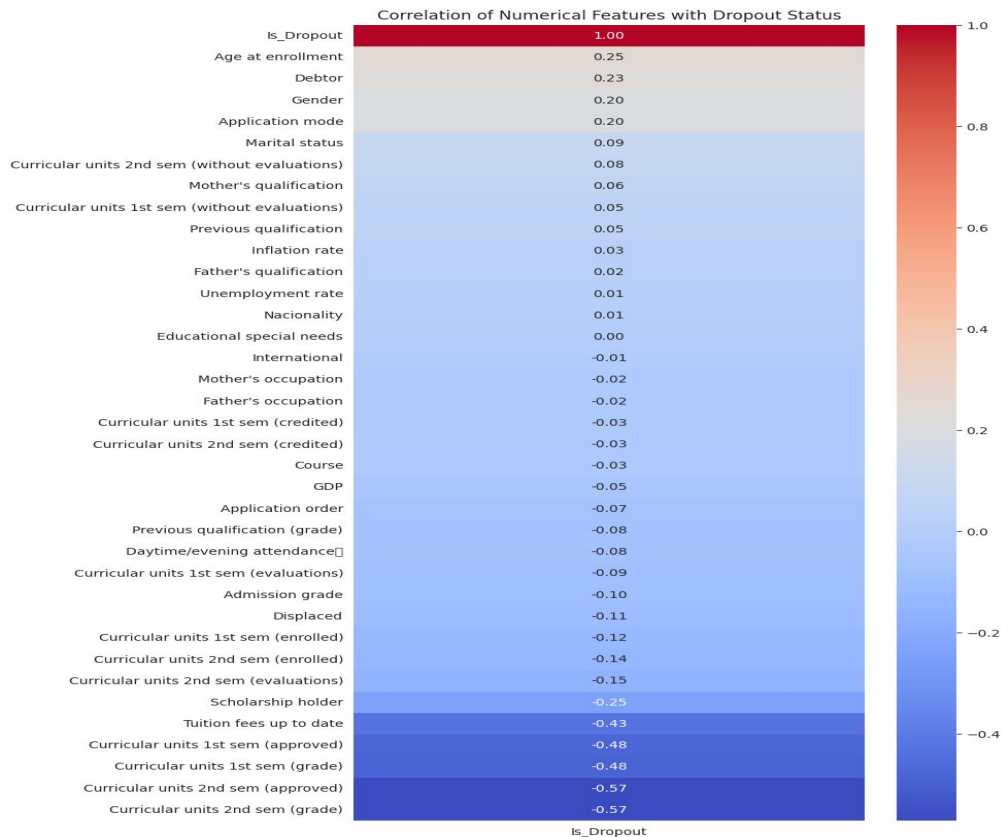
Demographic: age, prior schooling, course

Socioeconomic: tuition status, scholarship, financial aid

Note: no categorical data

Key challenge: the “Enrolled” group is the smallest (18%) and the hardest to predict. They look like both future graduates and future dropouts.

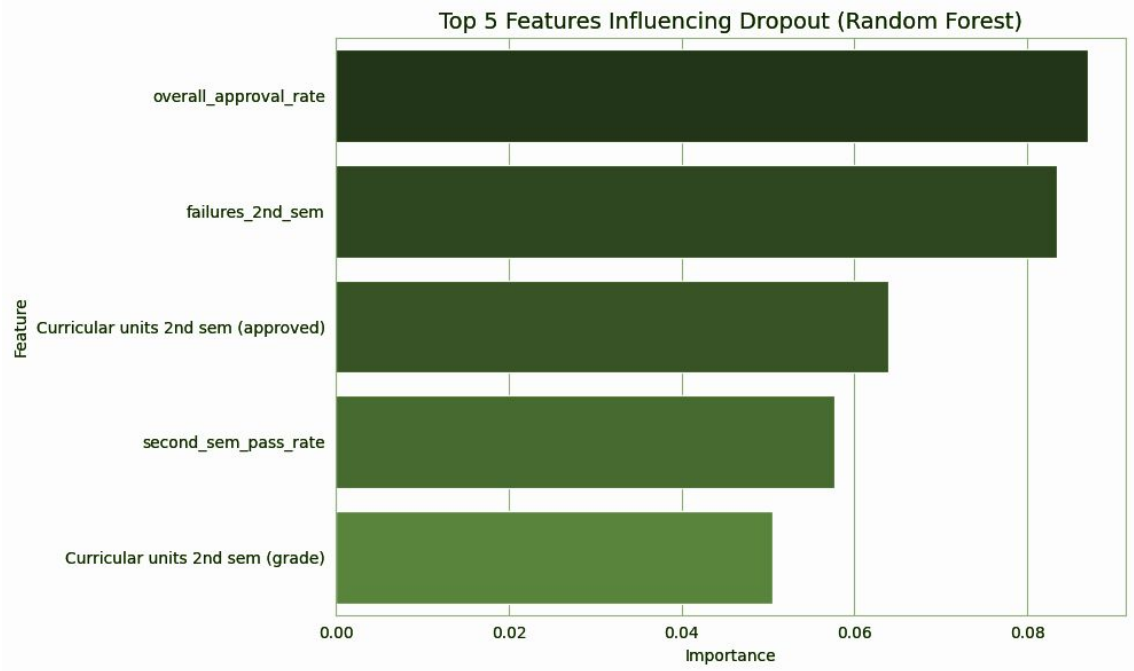
Academic signals stood out immediately



What we saw

- Students who pass their classes stay enrolled.
- Failing or skipping classes tracks closely with dropping out.
- Background factors mattered far less than performance.

What actually predicts dropout



The story is simple

The top 5 predictors are **all academic performance**: how many classes a student passes in their first year.

That means risk is **visible by the end of year 1**, while there's still time to act.

Our final modeling process

1

5+ models tested

Random Forest and XGBoost ended up performing the best

2

Train / test split

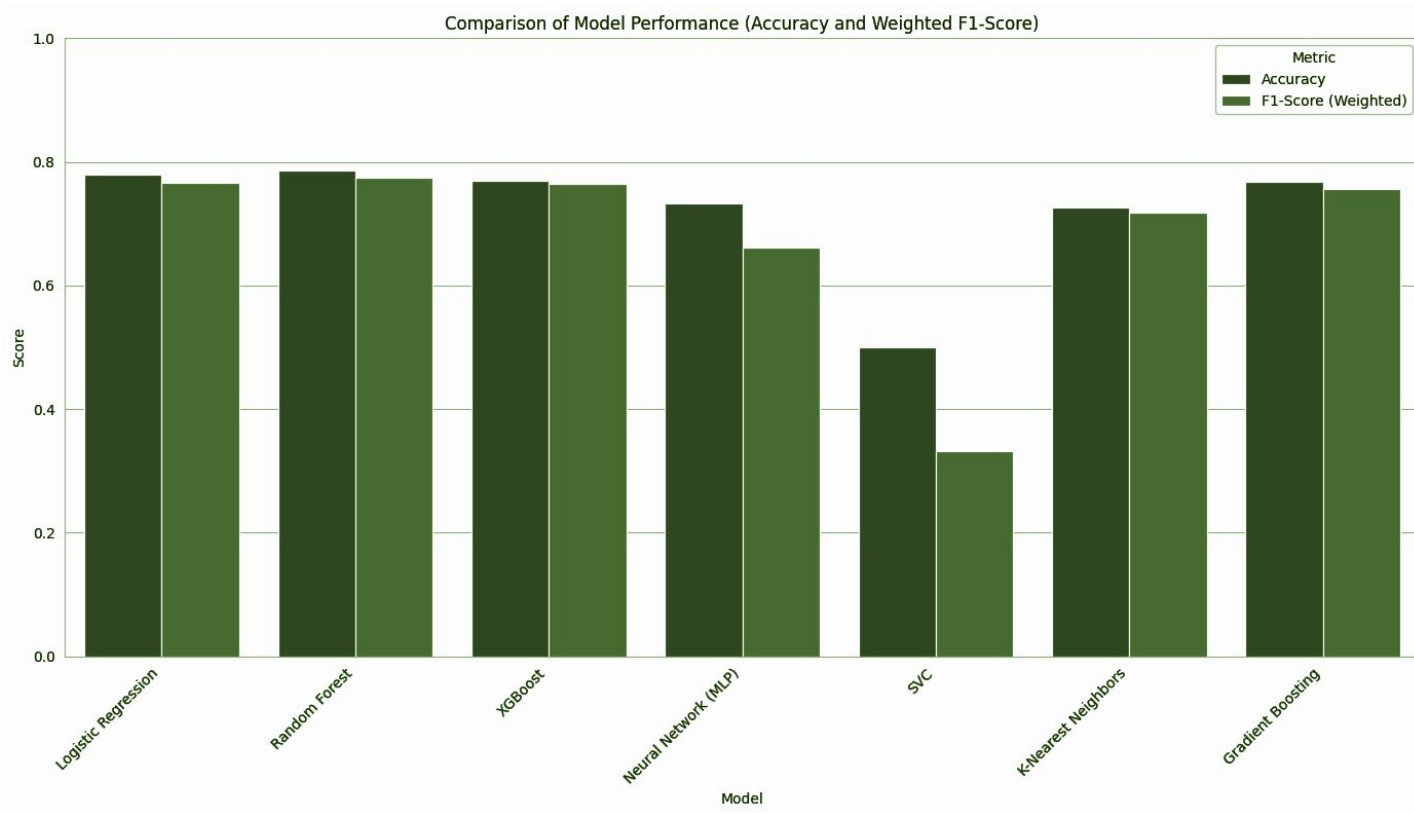
Trained on 80% of students; tested on 20%

3

Benchmark

We focused on how many at-risk students the model catches, not just its overall accuracy.

Comparison of Model Performance

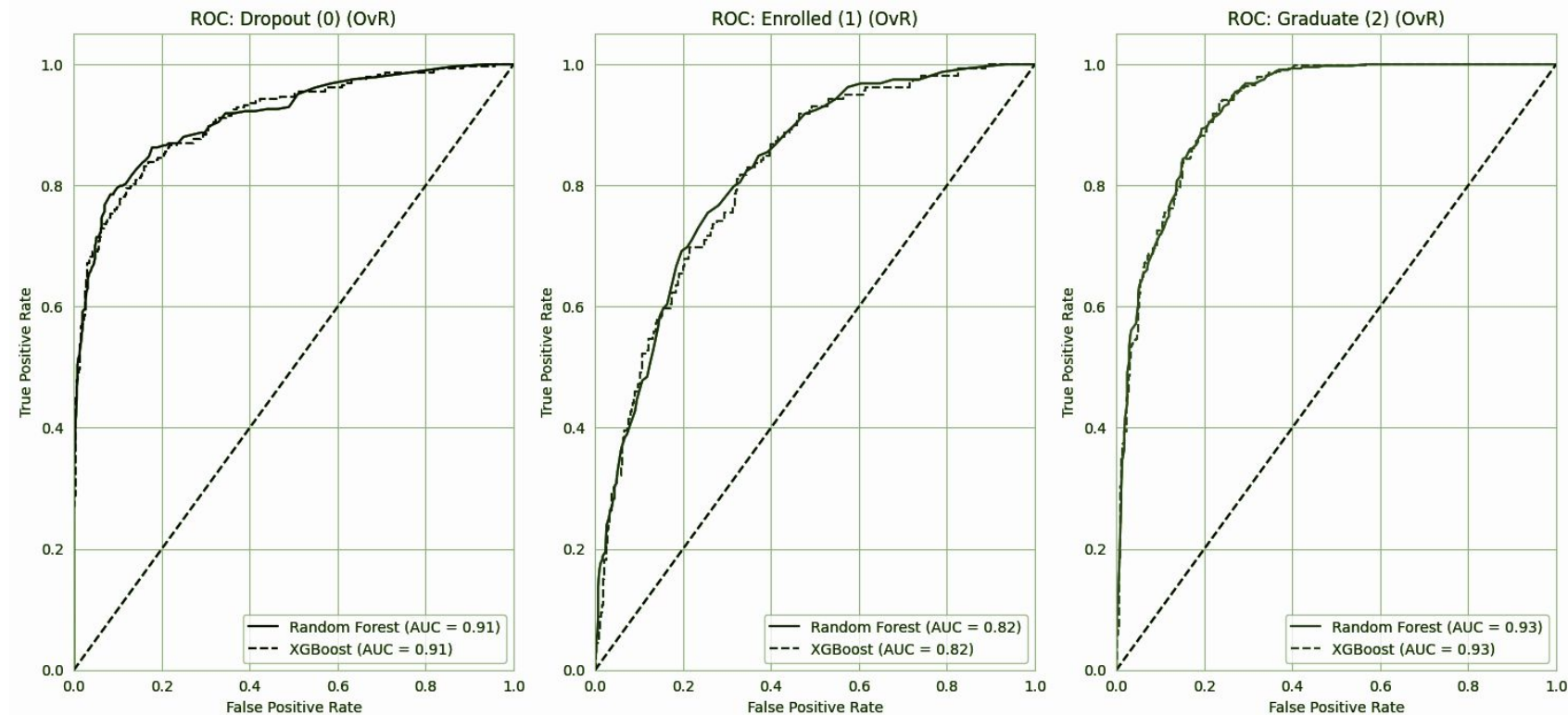


Random Forest won

Model	Accuracy	F1-score
Random Forest	77.4%	0.76
XGBoost	76.3%	0.75

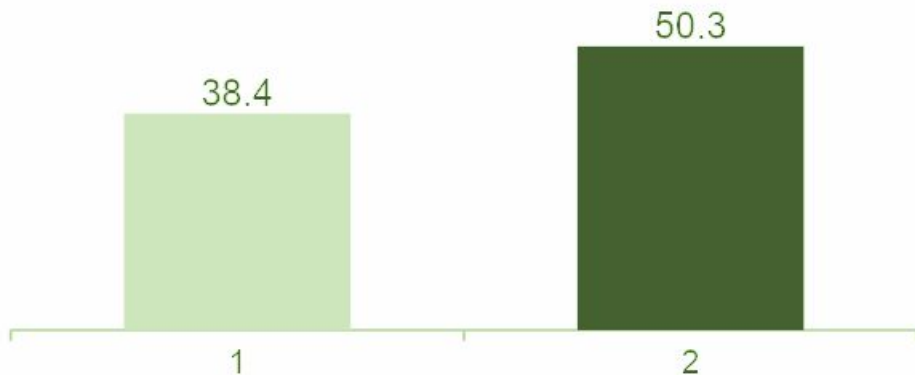
Why it's trustworthy: the ROC curves (next slide) show the model separates the classes well above random chance, and test performance held up on unseen students.

ROC Curves



Catching “Enrolled” students

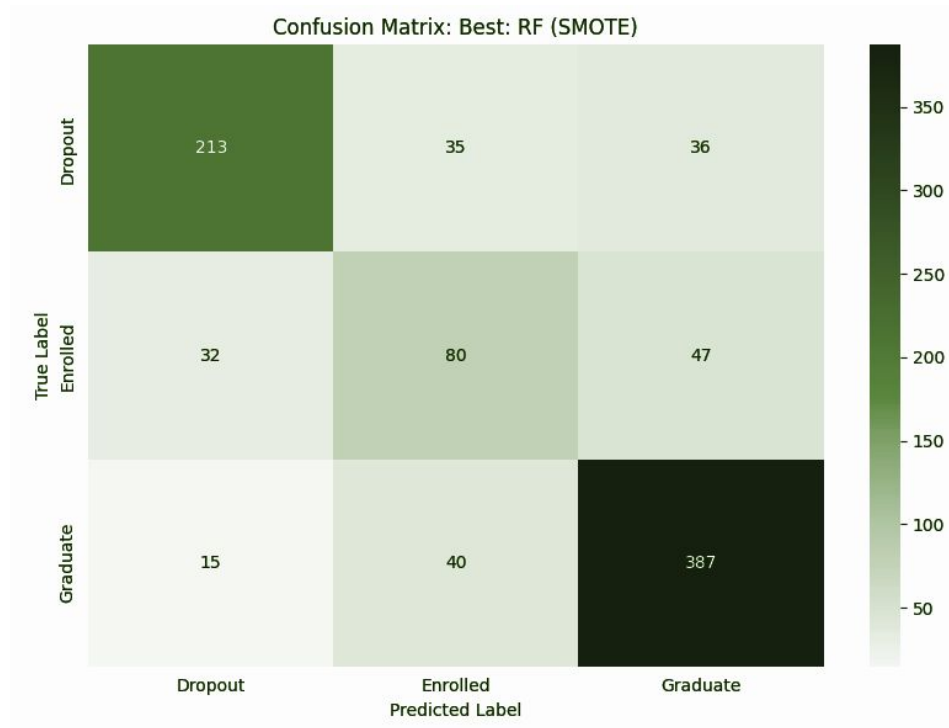
These students overlap both other groups, so any model struggles, and that's why 3-way accuracy caps near 77%. We rebalanced the training data (SMOTE) to catch more of them:



+12pts

We nearly doubled how often the model correctly catches at-risk Enrolled students at almost no cost to overall accuracy.

How well does it catch at-risk students?



Reading it plainly

Each row is what really happened; each column is the model's guess.

Dropouts and graduates are **caught reliably**.

Most mistakes are Enrolled students confused with one of the other two, which is exactly the overlap we expected.

Ask the easy question first

Instead of guessing all three outcomes at once, we first ask only: "Will this student drop out: yes or no?" That single decision is what triggers an intervention and it's easier to get right.

Catches 76% of real dropouts
before they leave

Tunable: we can dial sensitivity up to catch even more at-risk students when staff have capacity to follow up

87%

accurate at flagging dropout risk
(yes/no)

Three takeaways for the institution

Risk is visible early

Every top predictor is first-year academic performance. You don't need to wait to act.

The signal is academic

It's about passing classes, not background. So the fix is academic support, not guesswork.

A reliable yes/no flag

An 87%-accurate dropout flag is good enough to build a real early-warning watchlist on.

A 3-tier early-warning program

Tier 1

Flag early

Monitor first-semester pass rates and failures. Auto-flag low performers by mid-semester for mandatory tutoring and a personalized success plan.

Tier 2

Reinforce 2nd sem

Build on it: time-management & study workshops for all. One-on-one coaching, faculty mentorship, and peer support for the students still struggling.

Tier 3

Support holistically

A cross-department group (advising, mental health, financial aid) to address the non-academic barriers behind poor performance.

Risk is visible early. Now we can act on it.

We can flag at-risk students with 87% accuracy using only first-year academic signals. The signals are specific enough to know what kind of support helps and aid administrators in intervening quickly.

WHAT WE'D DO NEXT

- Test the dropout flag on more groups of students to confirm it holds up
- Try other models and give it more student information
- Run a real trial with an advising team to see if it helps students





***"Education is our passport to the future,
for tomorrow belongs to the people who
prepare for it today."***

— Malcolm X



Sources

UCI ML Repository — Predict Students' Dropout and Academic Success (Dataset 697)

Thank You

Questions?
